Abstract Algebra - Stockholm University 2025

1. Groups

1.1. Conjugation, centralizer, center, normalizer

For $g \in G$, the conjugation map $c_g(x) = gxg^{-1}$ is an automorphism of G (an *inner* automorphism). For a subset $S \subseteq G$, the centralizer is

$$C_G(S) = \{ g \in G : \ gs = sg \ \text{ for all } s \in S \}; \quad C_G(g) := C_G(\{g\}).$$

The center is

$$Z(G) = C_G(G) = \{g \in G : \ gx = xg \ \forall x \in G\}$$

(always a characteristic, hence normal, subgroup). For a subgroup $H \leq G$, the normalizer is

$$N_G(H) = \{ g \in G : \ gHg^{-1} = H \},$$

the largest subgroup of G in which H is normal (so $H \trianglelefteq N_G(H)$). One always has $C_G(H) \leq N_G(H)$. The number of conjugates of H in G equals $[G : N_G(H)]$.

1.2. Normal subgroups and quotients

A subgroup $N \leq G$ is normal (write $N \trianglelefteq G$) if $gNg^{-1} = N$ for all $g \in G$. Equivalently, every left coset equals the corresponding right coset: $gN = Ng$. If $N \trianglelefteq G$, the set of cosets $G/N = \{gN : g \in G\}$ is a group with $(gN)(hN) = (gh)N$.

1.3. Cosets and index

For $H \leq G$ and $g \in G$, the left (resp. right) coset is $gH = \{gh : h \in H\}$ (resp. $Hg = \{hg : h \in H\}$). Left cosets are either disjoint or equal and they partition G. Each left coset gH has $|H|$ elements via the bijection $H \to gH$, $h \mapsto gh$. The *index* of H in G is the number of left cosets, denoted $[G:H]$.

1.4. Lagrange's Theorem

If G is finite and $H \leq G$, then

$$|G| \ = \ [G:H] \cdot |H|.$$

Proof sketch. G is the disjoint union of $[G:H]$ left cosets, each of cardinality $|H|$.

Consequences

- $|H| \mid |G|$ for every $H \leq G$.
- For $g \in G$, the order $|g|$ divides $|G|$; hence $g^{|G|} = e$.
- Index multiplicativity: for $K \leq H \leq G$ (with G finite), $[G:K] = [G:H]\,[H:K]$.
- If $[G:H] = 2$, then $H \trianglelefteq G$ (there are only two cosets, so gHg^{-1} must be H).

Groups of prime order are cyclic. If $|G| = p$ (prime) and $g \in G$, $g \neq e$, then $|\langle g\rangle| = |g| \mid p$ by Lagrange, so $|g| = p$ and $\langle g\rangle = G$. Hence G is cyclic and $G \cong \mathbb{Z}/p\mathbb{Z}$.

1.5. Isomorphism theorems

(1) **First isomorphism theorem.** For $\varphi : G \to H$ a homomorphism,

$$G/\ker\varphi \;\cong\; \operatorname{im}\varphi.$$

(2) **Second isomorphism theorem.** For $A \leq G$ and $N \trianglelefteq G$,

$$A \cap N \trianglelefteq A, \qquad AN \leq G, \qquad AN/N \;\cong\; A/(A \cap N).$$

(3) **Third isomorphism theorem.** If $K \leq H \trianglelefteq G$, then $H/K \trianglelefteq G/K$ and

$$(G/K)/(H/K) \;\cong\; G/H.$$

(4) **Correspondence (lattice) theorem.** For $N \trianglelefteq G$, the map

$$\{\,\text{subgroups } H \mid N \leq H \leq G\,\} \;\longleftrightarrow\; \{\,\text{subgroups of } G/N\,\}, \quad H \mapsto H/N$$

is a bijection preserving inclusion and index; it carries normal subgroups $H \trianglelefteq G$ with $N \leq H$ to normal subgroups of G/N.

2. Group actions

2.1. Actions on cosets and by conjugation

- For $H \leq G$, the action $G \curvearrowright G/H$ by left multiplication is transitive, with stabilizer of aH equal to aHa^{-1}.
- The action $G \curvearrowright G$ by conjugation, $g \cdot x = gxg^{-1}$, partitions G into conjugacy classes. The stabilizer of x is its centralizer $C_G(x)$, and $|\mathrm{Orb}(x)| = [G : C_G(x)]$.

2.2. Cayley's Theorem

Every group G is isomorphic to a subgroup H of the symmetric group $S_{|G|}$:

$$G \cong H \leq S_{|G|}.$$

2.3. Orbit–stabilizer

For any action $G \curvearrowright X$ and any $x \in X$ there is a canonical bijection

$$G/G_x \xrightarrow{\sim} \mathrm{Orb}(x), \qquad gG_x \longmapsto g \cdot x.$$

In particular, if G is finite then

$$|\mathrm{Orb}(x)| = [G : G_x] \quad \text{and} \quad |G| = |\mathrm{Orb}(x)| \cdot |G_x|.$$

2.4. Class Equation

For a finite group G, let $Z(G)$ be the center of G. Then

$$|G| \;=\; |Z(G)| \;+\; \sum_{i=1}^{r} [G : C_G(g_i)],$$

where $g_1, \ldots, g_r$ are representatives of the conjugacy classes not contained in $Z(G)$.

2.5. Conjugacy in S_n

Two permutations in S_n are conjugate if and only if they have the same cycle type (i.e. the same cycle lengths with the same multiplicities).

$$\sigma, \tau \in S_n \quad \text{are conjugate} \quad \Longleftrightarrow \quad \text{cycle-type}(\sigma) = \text{cycle-type}(\tau).$$

Hence, the conjugacy classes of S_n correspond bijectively to partitions of n.

2.6. Centralizer size in S_n

If $\sigma \in S_n$ has, for each length $\ell \geq 1$, exactly m_ℓ disjoint cycles of length ℓ (so $\sum_{\ell \geq 1} \ell\, m_\ell = n$), then

$$|C_{S_n}(\sigma)| = \prod_{\ell \geq 1} \ell^{m_\ell}\, m_\ell!\,.$$

Here ℓ is a cycle length and m_ℓ is the number of ℓ-cycles in σ. Consequently, the conjugacy class size is

$$|\mathrm{Cl}(\sigma)| = \frac{n!}{|C_{S_n}(\sigma)|}.$$

2.7. Automorphisms of cyclic groups

For $n \geq 1$, the automorphism group of the cyclic group $C_n \cong \mathbb{Z}/n\mathbb{Z}$ is

$$\mathrm{Aut}(C_n) \;\cong\; (\mathbb{Z}/n\mathbb{Z})^\times,$$

hence $|\mathrm{Aut}(C_n)| = \varphi(n)$. Here $\varphi(n)$ is *Euler's totient function*:

$$\varphi(n) = \left|(\mathbb{Z}/n\mathbb{Z})^\times\right| = |\{\, 1 \leq k \leq n : \gcd(k, n) = 1 \,\}|.$$

Equivalently, if $n = \prod p_i^{e_i}$ then

$$\varphi(n) = n \prod_i \Big(1 - \frac{1}{p_i}\Big) = \prod_i \big(p_i^{e_i} - p_i^{e_i - 1}\big).$$

3. Sylows Theorem

Let $|G| = p^a m$ with $p \nmid m$ and $a \geq 1$. A subgroup of order p^a is a *Sylow p-subgroup.* Write $\mathrm{Syl}_p(G)$ for the set of Sylow p-subgroups and $n_p := |\mathrm{Syl}_p(G)|$.

3.1. Statements

- $\mathrm{Syl}_p(G) \neq \varnothing$.
- Every p-subgroup of G is contained in some conjugate of a Sylow p-subgroup. In particular, any two Sylow p-subgroups of G are conjugate in G.
- G acts on $\mathrm{Syl}_p(G)$ by $g \cdot P = gPg^{-1}$. For $P \in \mathrm{Syl}_p(G)$, the stabilizer is $N_G(P)$, hence
$$n_p = [G : N_G(P)].$$
- $n_p \equiv 1 \pmod p$ and $n_p \mid m$.
- If $n_p = 1$, then the unique Sylow p-subgroup P is normal *and characteristic* in G (automorphisms permute Sylow p-subgroups; uniqueness forces P to be fixed).

3.2. Fundamental Theorem of Finitely Generated Abelian Groups

Every finitely generated abelian group G decomposes uniquely (up to isomorphism) as
$$G \;\cong\; \mathbb{Z}^r \;\times\; \mathbb{Z}_{n_1} \times \cdots \times \mathbb{Z}_{n_s}, \quad \text{with } r \geq 0,\ n_j \geq 2, \text{ and } n_{i+1} \mid n_i.$$

Consequences. $G \cong \mathbb{Z}^r \times T(G)$ where $T(G)$ is the (finite) torsion subgroup. If $|G| = \prod p_i^{a_i}$ then the number of abelian groups of order $|G|$ (up to isomorphism) is
$$\prod_i p(a_i),$$
where $p(k)$ is the number of partitions of k.

Counting abelian groups To find the number of abelian groups of a given finite order:

1. Factor the order: $|G| = \prod p_i^{a_i}$.
2. For each prime p_i, list all partitions of a_i.
3. Multiply their counts: number of groups $= \prod_i p(a_i)$.

Example: $|G| = 60 = 2^2 \cdot 3 \cdot 5$ gives $p(2)p(1)p(1) = 2$ abelian groups: $\mathbb{Z}_{60}$ and $\mathbb{Z}_{30} \times \mathbb{Z}_2$.

3.3. Groups of order p^2

Let $|G| = p^2$ with p prime. Then G is abelian, and
$$G \cong C_{p^2} \quad \text{or} \quad G \cong C_p \times C_p.$$

Proof. Since G is a p-group, $Z(G) \neq 1$. If $|Z(G)| = p^2$ then $Z(G) = G$ and G is abelian. Otherwise $|Z(G)| = p$, so $|G/Z(G)| = p$ is cyclic, and a group with cyclic quotient by its center is abelian. The classification follows from the classification of finite abelian groups.

4. Rings

4.1. Ideals and basic ideal arithmetic

Throughout, R is a commutative ring with 1.

Ideal. An *ideal* $I \trianglelefteq R$ is an additive subgroup $I \le (R,+)$ with the absorption property: $rI \subseteq I$ for all $r \in R$ (equivalently $Ir \subseteq I$ in the commutative case).

Principal ideal. For $a \in R$, $(a) := Ra = \{ra : r \in R\}$.

Zero and unit ideals. $0 := \{0\}$ and R are ideals. Every ideal contains 0. $1 \in I$ *iff* $I = R$ (the *unit ideal*); otherwise $1 \notin I$.

Sum and intersection. For ideals $I, J \trianglelefteq R$,

$$I + J := \{\, i + j : \ i \in I,\ j \in J \,\}, \qquad I \cap J \trianglelefteq R.$$

$I + J$ is the *smallest* ideal containing $I \cup J$; $I \cap J$ is the *largest* ideal contained in both.

Product.

$$IJ := \Big\{ \ \sum_{k=1}^{m} i_k j_k : \ i_k \in I,\ j_k \in J,\ m \ge 1 \ \Big\}.$$

Always $IJ \subseteq I \cap J$. If $I + J = R$ (i.e. I, J are *comaximal*), then $IJ = I \cap J$.

Powers. $I^n := \underbrace{I \cdot I \cdots I}_{n \text{ times}}$; then $I^{n+1} \subseteq I^n$.

Quotients. R/I is a ring; the projection $\pi : R \to R/I$ is surjective with $\ker \pi = I$. Ideals J with $I \subseteq J \trianglelefteq R$ correspond to ideals of R/I via $J \mapsto J/I$.

Prime and maximal via quotients. For a proper ideal $I \trianglelefteq R$: I is prime $\iff$ R/I is an integral domain; I is maximal $\iff$ R/I is a field. In a PID, every nonzero prime ideal is maximal.

Polynomial rings over fields. If F is a field, then $F[x]$ is Euclidean (degree), hence a PID and a UFD. Units of $F[x]$ are $F^\times$.

4.2. Ideals generated by two polynomials in $F[x]$

Let F be a field and $f, g \in F[x]$. If $d = \gcd(f, g)$ is chosen monic, then

$$(f, g) = (d).$$

Reason. Since $F[x]$ is Euclidean, there exist $a, b \in F[x]$ with $af + bg = d$, so $(d) \subseteq (f, g)$. Conversely, $d \mid f$ and $d \mid g$, hence $(f, g) \subseteq (d)$. Thus $F[x]/(f, g) \cong F[x]/(d)$. In particular, in $\mathbb{Q}[x]$ one may always take d monic.

4.3. Chinese Remainder Theorem

Let R be a commutative ring with 1 and let $A_1, \dots, A_k \trianglelefteq R$ be ideals. If the A_i are pairwise comaximal ($A_i + A_j = R$ for $i \neq j$), then the natural map

$$\phi : R \longrightarrow \prod_{i=1}^{k} R/A_i, \qquad r \mapsto (r + A_i)_i$$

is surjective with kernel $\bigcap_{i=1}^{k} A_i = \prod_{i=1}^{k} A_i$. Hence

$$R/\big(\prod_{i=1}^{k} A_i\big) \;\cong\; \prod_{i=1}^{k} R/A_i.$$

5. Integral -, Unique Factorization - & Principle Ideal Domains

5.1. Definitions

Integral domain. A (commutative, unital) ring R with $1 \neq 0$ and no zero divisors.

Unique Factorization Domain (UFD). An integral domain in which every nonzero nonunit factors as a finite product of irreducibles, and this factorization is unique up to order and associates.

Principal Ideal Domain (PID). An integral domain in which every ideal is principal, i.e. $\forall I \trianglelefteq R$ there exists $a \in R$ with $I = (a)$.

Euclidean domain (ED). An integral domain R with a function $\delta : R \setminus \{0\} \to \mathbb{N}$ such that for all $a, b \in R$ with $b \neq 0$ there exist $q, r \in R$ with $a = bq + r$ and either $r = 0$ or $\delta(r) < \delta(b)$ (a "division algorithm").

Fields are Euclidean. For a field F, define $\delta(x) = 0$ for $x \neq 0$. Then for any $a, b \in F$ with $b \neq 0$, $a = b(ab^{-1}) + 0$, so F is Euclidean (hence a PID and a UFD).

Implications

$$\text{Field} \;\Rightarrow\; \text{Euclidean domain} \;\Rightarrow\; \text{PID} \;\Rightarrow\; \text{UFD} \;\Rightarrow\; \text{integral domain.}$$

5.2. Gauss's Lemma

Let R be a UFD with field of fractions F, and let $p(x) \in R[x]$. If p is reducible in $F[x]$, then p is reducible in $R[x]$. (Equivalently: if p is irreducible in $R[x]$, then it is irreducible in $F[x]$.)

5.3. Eisenstein's Criterion

Let R be an integral domain and $P \subset R$ a prime ideal. Consider

$$f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0 \in R[x].$$

If $a_{n-1}, \dots, a_0 \in P$ and $a_0 \notin P^2$, then $f(x)$ is irreducible in $R[x]$.

Corollary ($\mathbb{Z}[x]$). Let $f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0 \in \mathbb{Z}[x]$. If there exists a prime p such that $p \mid a_i$ for $i = 0, \ldots, n-1$ and $p^2 \nmid a_0$, then f is irreducible in $\mathbb{Z}[x]$ (hence in $\mathbb{Q}[x]$ by Gauss).

5.4. Rational Root Test

Let $f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0 \in \mathbb{Z}[x]$ with $a_n \neq 0$. If $\frac{p}{q} \in \mathbb{Q}$ (in lowest terms, i.e. $\gcd(p, q) = 1$) is a root of f, then

$$p \mid a_0 \qquad \text{and} \qquad q \mid a_n.$$

In particular, if f is *monic* ($a_n = 1$), every rational root must be an *integer* divisor of a_0.

6. Field Extensions

6.1. Simple extensions and minimal polynomials

If α is algebraic over a field F, there is a unique monic irreducible $m_\alpha(x) \in F[x]$ with $m_\alpha(\alpha) = 0$ (the *minimal polynomial* of α).

6.2. Construction via quotients

If $p(x) \in F[x]$ is irreducible and K/F contains a root α of p, then

$$F(\alpha) \;\cong\; F[x]/(p(x)).$$

In particular, for any irreducible p, the quotient $F[x]/(p)$ is a field of degree $\deg p$, and the class $\bar{x} := x + (p)$ is a root of p.

6.3. Degree and basis of a simple extension

Let α be algebraic over F with m_α of degree n. Then

$$[F(\alpha) : F] = n, \qquad \{1, \alpha, \alpha^2, \ldots, \alpha^{n-1}\} \text{ is an } F\text{-basis of } F(\alpha).$$

6.4. Quotients

Let F be a field and $0 \neq f \in F[x]$ be nonconstant. Then

$$F[x]/(f) \text{ is a field} \iff f \text{ is irreducible in } F[x].$$

If f is reducible in $F[x]$, then $F[x]/(f)$ is not a field (it has zero divisors).

6.5. Tower law

For fields $F \subseteq K \subseteq L$,

$$[L : F] = [L : K]\,[K : F].$$

6.6. Finite $\Rightarrow$ algebraic

If K/F is a finite extension, then every $\alpha \in K$ is algebraic over F (hence K/F is algebraic).